

2. Percentage of preparations

Objectives of the class

- **At the end of the class you will be able to:**
- Define the expressions *percent weight-in-volume*, *percent volume-in -volume*, and *percent weight-in-weight*
- Calculate the percentage strength of a pharmaceutical preparation
- Apply percent strength to calculate the quantity of an ingredient present in a pharmaceutical preparation

Percentage of preparations

(percentage W/V, V/V, W/W)

- ❑ **%w/v (percent weight in volume)** - expresses the number of **grams** of a drug in **100 ml of liquid preparation**.

$$\% \frac{w}{v} = \frac{\text{weight of a drug (g)}}{\text{volume of a preparation (mL)}} \times 100$$

- ❑ **%v/v (percent volume in volume)** - expresses the number of **milliliters** of a drug in **100 ml** of liquid preparation.

$$\% \frac{v}{v} = \frac{\text{Volume of a drug (mL)}}{\text{Volume of a preparation (mL)}} \times 100$$

Percentage preparations (cont'd)

- ❑ **%w/w (percent weight in weight)**- expresses the number of **grams** of a drug in **100 g** of preparation or mixture.

$$\% \frac{w}{w} = \frac{\text{Weight of a drug (g)}}{\text{weight of a preparation (g)}} \times 100$$

- ❑ In general, the nature of ingredients in a pharmaceutical preparation determines the basis of calculation.

Percentage preparations (cont'd)

- ✓ A solid drug dissolved or suspended in a liquid vehicle would generally be calculated on **weight-in-volume** basis.
- ✓ A solid drug mixed with a solid or semisolid would generally be calculated on a **weight-in-weight** basis.
- ✓ A liquid drug in a liquid preparation would be calculated on **volume-in-volume** basis.
-  *Care is needed in making a % v/v solution, since the volume of a mixture may not be equal to the sum of the individual volumes.*

Examples

1. If 200 g of dextrose is used in preparation of 4000 mL of dextrose solution, what will be the concentration of dextrose (%w/v) in the solution? Ans: 5%
2. How many milliliters of a 3% solution can be made from 27 g of ephedrine sulfate? Ans: 900 ml
3. How many grams of a drug are required to prepare 120 ml of 12.5% aqueous solution? Ans: 15 g

Examples...

4. What is the percentage strength v/v of a solution of 800 g of a liquid with a specific gravity of 0.800 in enough water to make 4000 mL? Ans: 25% (v/v)
4. How many liters of a mouthwash can be prepared from 100 mL of cinnamon flavor if its concentration is to be 0.5% (v/v)? Ans: 20 L
4. The solubility of boric acid is 1 g in 18 mL of water at 25 °C. What is the percentage strength, w/w, of a saturated solution? Ans: 5.26%

Examples...

7. How many grams of sucrose must be added to 475 mL of water to make a 65% (w/w) solution?

Ans: 882.14 g

8. A metronidazole vaginal gel contains 0.75% of drug in 70-g tubes. An applicator will hold 5 g of gel for each administration. How much drug will be contained in each application? Ans: 37.5 mg

Ratio strength

- The concentrations of **dilute solutions** are frequently expressed in terms of ratio strength.

For example, 5% means *5 parts per 100* or
 $5:100 = 1:20$

- Express 0.02% to ratio strength. Ans: 1:5000
- Express (1/8)% to ratio strength.

Ratio strength...

- If the ratio, 1:1000 is given, it can be interpreted in the following manner:
 - ✓ ***For solids in liquids:*** 1 g of solute or constituent in 1000 mL of solution or liquid preparation.
 - ✓ ***For liquids in liquids:*** 1 mL of constituent in 1000 mL of solution or liquid preparation.
 - ✓ ***For solids in solids:*** 1 g of constituent in 1000 g of mixture.

Examples

1. A cream contains 0.01% (w/v) of dienestrol. Express this concentration as a ratio strength.
2. An ophthalmic solution is preserved with 0.0008% (w/v) of an antibacterial. Express this concentration as a ratio strength.

Conversions of concentration to mg/mL

- Occasionally, pharmacists, particularly those practicing in patient care settings, need to *convert rapidly* product concentrations expressed as percentage strength in w/v, ratio strength, or as grams per liter (as in IV infusions) to milligrams per milliliter (mg/mL).
 - To convert product percentage strength in % w/v to mg/mL, multiply the percentage strength expressed by 10.

Conversions of concentration to mg/mL (cont'd)

- To convert product **ratio strength to mg/ml**, divide the **denominator of ratio strength** by **1000**.
- The final conversion may be written in 1 mg/ **x mL** form. Where **x = Denominator ÷ 1000**.
- To convert product strengths expressed as **grams per liter (g/L) to mg/ml**, convert the numerator to milligrams, denominator to milliliters and then divide the number of milligrams in the numerator by the number of milliliters in the denominator.

Parts per million (ppm)

- Concentrations of **very dilute preparations** are also expressed in terms of **parts per million (ppm)**, i.e., the number of parts of a drug per million parts of a whole.
- ✓ A 1 ppm solution contains 1 part of drug per 1 million parts of solution.
- ✓ A 5 ppm is 5 parts of drug per 1 million parts of solution.

Parts per million (ppm)

- When ppm is used as a designation for concentration, some systems are w/w, some are v/v, and some are w/v.
- For w/w and v/v cases the two units must be the same.
- Weight by volume (w/v) concentrations are always defined as weight is in **grams** and volume is in **milliliters**.

Parts per million (ppm)...

- To calculate concentration in terms of **ppm**, we can use the following formula:

$$\text{ppm} = \frac{\text{weight or volume of solute}}{\text{weight or volume of solution or mixture}} \times 1,000,000$$

Examples

1. If 0.175 mg of a preservative presents in 5 mL of a syrup, what will be its concentration in ppm? Ans: 35 ppm
2. If a city water supply has a limit of 250 ppm of nitrate ion, what is the maximum amount of nitrate ion, in grams, that may be present in a 10,000-gallon reservoir? (1 gallon = 3785 mL). Ans: 9462.5 g

3. Calculation of doses and formulas

Objectives

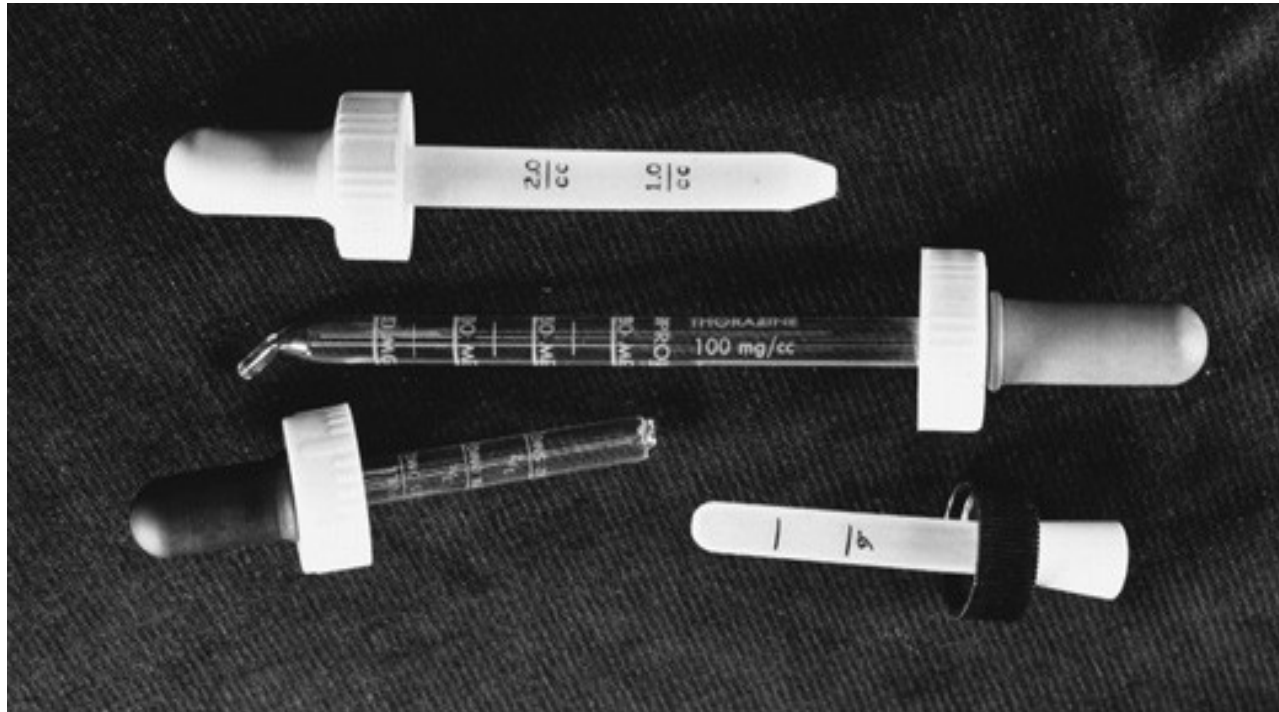
- At the end of the class you will be able to:
 - ✓ calibrate of medicine droppers
 - ✓ calculate number of doses and total quantity of medication
 - ✓ calculate doses based on age, weight and surface area
 - ✓ enlarge and reduce formulas

3.1 Calibration of droppers

- Medicine droppers are tools usually made from glass or plastic with a flexible rubber handle that is used to suction liquids in and out of the device.
- The most common use of a medicine dropper is to **deliver small amounts of medicine**.
- Most medicine droppers have a capacity of **1 to 2 mL**, making them ideal for administering medicine to infants and small children who don't require large doses.
- They can be also used to deliver/administer drops into eyes and ears of patients.

Medicinal droppers

❑ *Medicinal droppers*



Calibration of droppers.....(cont'd)

- The volume of a drop of liquid is different for different liquids.
- The size of drops of liquids varies with the **viscosity**, **surface tension**, and **density of the liquids**.
- Even two individuals dispensing the same liquid from identical droppers may produce drops of different sizes because of variations in the **pressure**, **speed of dropping**, and the **angle** at which the dropper is held.

Calibration of droppers.....(cont'd)

- Therefore, when distributing medication, it is very important to use the supplied dropper that came with the medicine.
- These droppers have been specifically calibrated for that particular medicine.

Calibration of droppers.....(cont'd)

- In order to maximize accuracy of dosing of compounded solution from a medicine dropper, the pharmacist must calibrate the dropper with the intended solution at the time of delivery.
- A dropper may be calibrated by counting the drops of a liquid as they fall in to a graduated cylinder until a measurable volume is obtained. The number of drops per unit volume is then established (e.g., 20 drops/ml).

Examples

1. If a pharmacist counted 40 drops of a medication in filling a graduate cylinder to the 2.5-mL mark, how many drops per milliliter did the dropper deliver? Ans: 16 drops/ ml

3.2 Calculation of doses

- **Dose** is the amount of drug taken at any one time. This can be expressed as the weight of drug (e.g. 250 mg of tablet), volume of solution of drug (e.g., 10 mL, 2 drops, etc).
- **Total quantity** is the total amount of drug taken during the course of therapy.
- **Dosage regimen** is the schedule of dosing. For example, four times per day for 10 days.

General dose calculations

- A pharmacist often needs to calculate the size of a dose, the number of doses, or the total quantity of medication to be dispensed.
- For these calculations the following equation is useful with the terms rearranged depending on the answer required.

General dose calculations.....(cont'd)

- In using the equation, the units of weight or volume must be the same for the total quantity and size of the dose.

$$\text{number of doses} = \frac{\text{total quantity}}{\text{size of dose}}$$

Examples

1. If the dose of a drug is 50 μg , how many doses are contained in 0.020 g? Ans: 400 doses.
2. A physician ordered 500-mg capsules of tetracycline to be taken twice a day for 10 days. How many total grams of tetracycline would be prescribed? Ans: 10 g

Examples...

3. The following regimen for oral prednisone is prescribed for a patient: 50 mg/day x 10 days; 25 mg/day x 10 days; 12.5 mg/ml x 10 days; 5 mg/day x 10 weeks. How many scored 25-mg tablets and how many 5-mg tablets should be dispensed to the dosing requirements?

Examples...

4. The recommended dose of memantine HCl is:

- Week 1, 5 mg/day
- Week 2, 10 mg/day (5 mg, b.i.d.)
- Week 3, 15 mg/day (10 mg a.m., 5 mg p.m.)
- Week 4, 20 mg/day (10 mg b.i.d.)

5. The formula for AURALGAN, an otic solution is:

Antipyrine.....54 mg

Benzocaine.....14 mg

Glycerin, dehydrated *ad*..... 1 mL

If a dropper delivers 20 drops/mL, how many milligrams of benzocaine would be delivered by a 3-drop dose of the solution?

Examples...

6. Rx

Acetaminophen Drops

Disp. 15 mL

Sig. 0.5 mL t.i.d.

- (a) If acetaminophen drops contain 1.5 g of acetaminophen per 15-mL container, how many milligrams are there in each prescribed dose?
- (b) If the dropper is calibrated to deliver 22 drops/mL, how many drops should be administered per dose?

Drug Dosage based on Body Weight

- In some cases, *the usual dose of drug* is expressed as a specific quantity of drug per unit of patient weight, such as milligrams of drug per kilogram of body weight (abbreviated mg/kg).
- The patient's weight is an important factor in dosing since the size of the body influences the drug's concentration in the body fluids and at its site of action.

Drug Dosage based on Body Weight.....(cont'd)

$$\text{Dose for a patient} = \frac{\text{dose of drug}}{\text{kg body weight}} \times \text{body weight (kg)}$$

Examples

1. The usual initial dose of chlorambucil is 150 $\mu\text{g}/\text{kg}$ of body weight. How many milligrams should be administered to a person weighing 154 lb.? (1 lb = 454.5 g)
2. A patient weighing 120 lb. was administered 2.1 g of a drug supposed to be dosed at 30 mg/kg. Was the dose administered *correct*, or was it an *overdose*, or was it an *underdose*?

Examples...

3. The adult dose of a liquid medication is 0.1 mL/kg of body weight to be administered as a single dose. How many spoonfuls should be administered to a person weighing 220 lb.? Ans:
2 tsp

Examples...

4. R_x

Erythromycin ethylsuccinate....400 mg/5 mL

Disp. 100 mL

Sig. _____ tsp. q.i.d. until all medication is taken.

If the dose of erythromycin ethylsuccinate is given as 40 mg/kg per day,

- (a) What would be the proper dose of the medication in the Signa if the prescription is for a 44-lb. child?
- (b) How many days will the medication last?

Drug Dosage based on Body Surface Area (BSA)

- Body surface area is recommended as the principal basis for drug dosage as many physiological functions are proportional to body surface area, such as metabolic rate and kidney function.
- In addition to the use of nomogram, BSA may be determined through use of the following formula:

Drug Dosage based on Body Surface Area (BSA)

- $$BSA (m^2) = \sqrt{\frac{Ht(cm) \times Wt(kg)}{3600}}$$

- $$Dose = \frac{Dose\ of\ drug}{m^2\ body\ surface\ area} \times Body\ surface\ area$$

Enlarging and reducing formulas

- Quantities of ingredients in pharmacopoeias such as USP/NF generally are determined based on 1000 mL or 1000 g of preparations.
- Those in the literature may be based on the preparation of a dosage unit (e.g., 5 mL, 1 capsule).
- Industrial formulas may be scaled up to quantities of ingredients sufficient to prepare hundreds of thousands of dosage units in a production batch.

Enlarging and reducing formulas.....(cont'd)

- Calculations to reduce or enlarge formulas may be performed by a two-step process.
- **Step 1**. Using the following equation, ***determine the factor*** that defines the multiple or the decimal fraction of the amount of formula to be prepared:

$$\text{Factor} = \frac{\text{Quantity desired}}{\text{Initial quantity in the formula}}$$

Enlarging and reducing formulas.....(cont'd)

- **Step 2.** Multiply the quantity of each ingredient in the formula by the factor to determine the amount of each ingredient required in the reduced or enlarged formula.

Examples 1. From the following formula, calculate the quantity of each ingredient required to make 240 mL of calamine lotion.

Calamine.....	80 g
Zinc Oxide.....	80 g
Glycerin.....	20 g
Bentonite Magma.....	250 mL
Calcium Hydroxide Topical Solution, to make.....	1000 mL

Examples...

2. From the following formula for 40 sertraline capsules, calculate the quantity of each ingredient that would be used if a pharmacist wished to prepare 250 capsules.

Sertraline Hydrochloride.....300 mg

Silica Gel.....6 g

Calcium Citrate..... 4 g

Examples...

3. The formula for an ophthalmic ointment is as follows:

Sulfacetamide sodium.....10% w/w

Prednisolone acetate.....0.2% w/w

Phenylmercuric acetate.....0.0008%w/w

Mineral Oil.....1% w/w

White Petrolatum, qs ad.....3.5 g

How much of each ingredient would be needed to manufacture 2000 such tubes of ointment?